FairFake: Systematic Demographic Bias Auditing and Explainable Visual Forensics for Deepfake Detection Systems

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***Abstract*—Although deep neural networks achieve remarkable benchmark accuracy in discriminating synthetic face manipulations from genuine media, emerging evidence reveals that these detectors exhibit acute performance disparities across demographic subgroups and incidental facial characteristics. In this investigation, we introduce FairFake, an integrated forensic auditing and visual attribution framework designed to measure, analyze, and interpret bias in facial manipulation detectors. We evaluate three prominent forensic architectures—Xception, EfficientNet-B0, and a Dual-Stream spatial-spectral network—across comprehensive multi-attribute annotations comprising 47 demographic, morphological, and cosmetic features. To prevent empirical artifacts stemming from skewed sample distributions, we formulate a control-group resampling audit methodology that decouples data imbalance from genuine model disparity through Corrected Relative Performance (*CRP*), Pristine Data Relative Performance (*PDRP*), and Deepfake Data Relative Performance (*DDRP*). Our extensive empirical audits reveal substantial systemic vulnerabilities: while macro-level accuracy remains high (up to 90.10% accuracy, 0.9570 AUC-ROC on N = 2,000 test portraits), detectors incur acute error amplification for subjects with darker complexions (*CRP* = +0.55), heavy cosmetics (*CRP* = +0.48), eyewear (*CRP* = +0.45), and baldness (*CRP* = +0.52). Through Gradient-weighted Class Activation Mapping (Grad-CAM) coupled with DeepFace biometric parsing, we show that convolutional kernels routinely latch onto cosmetic blending borders, specular glasses glare, and smiling facial creases as spurious cues of digital synthesis, resulting in severe false alarms on unmanipulated media and enabling manipulated media from minority cohorts to evade detection. Finally, perturbation stress testing reveals that environmental degradations (e.g., lossy compression and underexposure) exacerbate demographic disparities by up to ΔCRP = +0.21. We open-source our complete pipeline, benchmark suites, and interactive auditing application to advance equitable, transparent, and legally defensible multimedia forensics.**

***Index Terms*—Deepfake forensics, algorithmic fairness, demographic disparity, explainable artificial intelligence, Grad-CAM attribution, convolutional neural networks, media verification.**

I. INTRODUCTION

THE accelerating democratization of generative artificial intelligence has fundamentally altered the threat landscape of digital media integrity. Contemporary deep learning architectures—most notably conditional generative adversarial networks (cGANs), variational autoencoders (VAEs), and latent diffusion models—now synthesize hyper-realistic facial replacements and synthetic expressions with extraordinary visual plausibility [1], [2]. Although these breakthroughs enable compelling applications across entertainment and virtual telepresence, their weaponization in disinformation campaigns, non-consensual defamatory media, and automated financial deception presents profound challenges to civil society, democratic institutions, and jurisdictional evidentiary standards [3], [4].

In response, the computer vision and multimedia security communities have engineered an array of sophisticated detection mechanisms designed to identify invisible mathematical traces left by generative algorithms [5], [6]. State-of-the-art forensic classifiers typically leverage deep convolutional backbones—such as depthwise separable networks (e.g., Xception) and compound-scaled architectures (e.g., EfficientNet)—to spot microscopic pixel inconsistencies, blending boundaries, and high-frequency spectral irregularities [7], [8]. While these classifiers regularly surpass 90% accuracy on curated benchmark splits, a critical question of algorithmic equity has emerged: *Do deepfake detectors perform with equal reliability across diverse demographic groups and human appearance variations?*

A. Problem Formulation & Ethical Imperatives

Recent scholarship has begun highlighting that forensic computer vision models frequently inherit the pervasive demographic skews endemic to contemporary web-scraped face databases [9], [10]. When a facial manipulation detector is trained predominantly on narrow demographic profiles, its decision boundary becomes calibrated around specific baseline textures, skin tones, and facial contours. Consequently, authentic imagery from underrepresented groups—such as individuals with deeper skin pigmentation, distinctive facial features, or cultural headwear—is disproportionately vulnerable to erroneous classification as synthetic manipulation [11].

The societal ramifications of such asymmetric classification errors are severe and twofold:

1) False Accusation Risk (Civil Disparities): In legal, corporate, and social media moderation contexts, an inflated False Positive Rate (FPR) unfairly subjects innocent individuals belonging to marginalized cohorts to unfounded scrutiny, account suspension, or reputational damage [12].

2) Evasion Vulnerability (Security Asymmetries): Conversely, an inflated False Negative Rate (FNR) on specific demographic subgroups allows malicious actors to systematically disguise synthetic media featuring targeted identities, exploiting algorithmic blind spots to bypass automated forensic gateways [13].

Despite the gravity of these concerns, existing literature has largely treated deepfake detection as a monolithic classification task. Few investigations have rigorously decomposed the mechanistic factors underlying algorithmic bias across both demographic (e.g., age, gender, ethnicity) and non-demographic attributes (e.g., eyewear, cosmetics, hairstyles), and virtually none have paired statistical disparity measurement with transparent, visual attribution explanations.

B. Core Contributions of FairFake

To resolve these challenges, this paper presents **FairFake**, an explainable, end-to-end framework for auditing demographic disparities and understanding failure heuristics in automated face forgery detection. The principal contributions of this work are summarized as follows:

1. **Standardized Multi-Dataset Disparity Benchmarking:** We evaluate five prominent face forgery corpora (Celeb-DF [14], FF++ [15], DFD [16], DFDC [17], and DF-1.0 [18]) mapped against comprehensive multi-attribute annotations covering 47 demographic, morphological, and cosmetic features, providing an empirical baseline of representation skews.
2. **Mathematically Controlled Fairness Metrics:** We formulate a rigorous auditing methodology featuring control-group resampling to calculate Corrected Relative Performance (*CRP*), Pristine Data Relative Performance (*PDRP*), and Deepfake Data Relative Performance (*DDRP*), decoupling test-set distribution artifacts from genuine algorithmic biases.
3. **Comprehensive Multi-Model Forensic Auditing:** We conduct systematic audits across three widely deployed forensic backbones (Xception, EfficientNet-B0, and a Dual-Stream spatial-spectral architecture), quantifying error disparities across 31 appearance attributes.
4. **Explainable Attribution & Failure Mode Discovery:** We integrate Gradient-weighted Class Activation Mapping (Grad-CAM) with DeepFace biometric parsing, uncovering how convolutional filters erroneously treat natural cosmetic gradients, eyewear reflections, and facial wrinkles as digital manipulation artifacts.
5. **Environmental Robustness Stress Testing:** We quantify the degradation of both discriminative accuracy and demographic equity under operational perturbations, showing that lossy compression and underexposure compound algorithmic bias by up to $\Delta CRP = +0.21$.
6. **Open-Source Auditing Suite & Web Platform:** We release the complete FairFake software suite, combining an asynchronous FastAPI verification engine with a modern Next.js interactive interface for real-time forensic auditing and compliance reporting.

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Project software, interactive auditing interfaces, and benchmark evaluation suites are publicly accessible via GitHub: https://github.com/code-manush/FairFake.

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II. Related Work & Theoretical Background

A. Spatial and Frequency-Domain Media Forensics

Automated deepfake detection methodologies have evolved through two primary paradigms: spatial feature analysis and frequency-domain decomposition [5]. Spatial approaches leverage deep convolutional neural networks to capture subtle boundary inconsistencies, color blending seams, and unnatural textural artifacts introduced during facial donor-target warping [6], [7]. Early architectures such as MesoNet [19] demonstrated the efficacy of shallow networks, while contemporary systems predominantly utilize depthwise separable convolutions (Xception [20]) and efficient mobile bottlenecks (EfficientNet [21]) to discern pixel-level anomalies.

In parallel, frequency-domain forensics emerged to exploit structural weaknesses in upsampling and generative deconvolution operations [8]. Upsampling layers within generative autoencoders frequently generate periodic checkerboard artifacts visible in discrete Fourier transform (DFT) or discrete cosine transform (DCT) representations [22]. Dual-stream architectures capitalize on both paradigms by fusing spatial feature maps with 2D-FFT spectral magnitudes, providing enhanced robustness against post-processing compression [23].

B. Representation Disparities in Deepfake Databases

The training and benchmarking of forensic models rely heavily on large-scale public video datasets. As outlined in Table II, early benchmarks like DeepfakeTIMIT [24] and FFW [25] possessed limited subject diversity. Subsequent benchmarks—such as FaceForensics++ (FF++) [15], Google DeepFakeDetection (DFD) [16], Celeb-DF [14], and the Deepfake Detection Challenge (DFDC) [17]—vastly expanded video volume. However, as demonstrated in our distribution analysis (Section IV-A), these collections suffer from acute demographic skews, with fair-skinned identities constituting over 88% of popular corpora while African and Asian cohorts remain starkly underrepresented.

TABLE I
COMPARATIVE TAXONOMY OF ALGORITHMIC FAIRNESS AUDITS IN FACIAL FORGERY FORENSICS

Investigation	Attributes Audited	Models Benchmarked	Datasets Analyzed	Visual Attribution
Hazirbas *et al.* [9]	4 (Demo only)	2	1 (CC)	None
Trinh & Liu [10]	2 (Gender/Race)	3	1 (FF++)	None
Pu *et al.* [11]	2 (Gender/Skin)	1	1 (FF++)	None
Xu *et al.* [13]	47 (Demo/Non-demo)	3	4	Statistical Only
FairFake (Our Work)	**47 (Demo + Cosmetic)**	**3 + Dual-Stream**	**5 Benchmarks**	**Grad-CAM + Biometrics**

C. Bias Auditing and Explainable Attribution

Prior studies investigating fairness in face analysis systems have observed that convolutional filters inadvertently encode secondary demographic traits even when trained exclusively for biometric verification [26], [27]. In media forensics, Hazirbas *et al.* [9] evaluated five commercial detectors on the Casual Conversations dataset, identifying systematic error inflation on darker skin tones. Trinh and Liu [10] and Pu *et al.* [11] confirmed racial and gender performance gaps on subsets of FaceForensics++. However, as summarized in Table I, prior investigations were restricted to small attribute sets and lacked explainable attribution tools. FairFake addresses these deficiencies by coupling comprehensive 47-attribute statistical audits with Grad-CAM visual heatmaps, rendering detector biases both quantifiable and interpretable.

III. THE FAIRFAKE AUDITING FRAMEWORK

A. Architectural Overview

As depicted in Fig. 6, the FairFake framework organizes media processing into three coordinated pipelines:

1) Multimodal Forensic Scoring: Input face crops $\mathbf{x} \in \mathbb{R}^{H\times W\times 3}$ are preprocessed through spatial normalization and fed into candidate forensic backbones (Xception, EfficientNet-B0, or Dual-Stream RGB+FFT) to produce a calibrated posterior probability of digital manipulation $p_{\text{fake}} = \sigma(f_\theta(\mathbf{x}))$.

2) Explainable Grad-CAM Attribution: To identify the spatial regions dictating model decisions, we compute class-specific gradient maps flowing into the terminal convolutional feature map A^k. Neuron importance weights α_k are computed across spatial dimensions $U \times V$:

$$\alpha_k = \frac{1}{U\times V}\sum_{i=1}^{U}\sum_{j=1}^{V}\frac{\partial y^c}{\partial A_{ij}^k} \tag{1}$$

The visual localization map $L_{\text{Grad-CAM}}$ is obtained via a rectified linear combination of feature maps:

$$L_{\text{Grad-CAM}} = \text{ReLU}\sum_k \alpha_k A^k \tag{2}$$

which is subsequently upsampled to input resolution and color-mapped to isolate pixel-level attention triggers.

3) Biometric Demographic Parsing: Simultaneously, input faces are parsed via DeepFace neural classifiers to extract perceived demographic indicators (apparent age cohort, perceived gender, ethnic phenotype) and facial landmarks, synchronizing visual heatmaps with specific attribute tags.

B. Mathematical Formulation of Resampled Fairness Metrics

Evaluating algorithmic bias in media forensics requires addressing a fundamental statistical challenge: *unbalanced attribute distributions in testing corpora confound raw subgroup error rates*. If subjects exhibiting an attribute a (e.g., eyeglasses) happen to contain a higher proportion of compressed or difficult videos in the test split, naive error rate comparisons will falsely attribute dataset-level difficulty to algorithmic bias.

To resolve this confounding factor, FairFake deploys a control-group resampling strategy [13], [26]. For any attribute $a \in A$, we define the positive subset $S^{(+)}(a) = \{i : a_i = 1\}$ and negative subset $S^{(-)}(a) = \{i : a_i = 0\}$. We compute the sample size threshold N:

$$N = \min\left(|S^{(+)}(a)|, |S^{(-)}(a)|\right) \tag{3}$$

and draw balanced control groups $C^{(+)}(a)$ and $C^{(-)}(a)$ of cardinality N sampled uniformly from the testing corpus. We define the baseline Relative Performance (RP) on any data partition:

$$RP_{\text{type}}(a) = 1 - \frac{err_{\text{type}}^{(+)}(a)}{err_{\text{type}}^{(-)}(a)} \tag{4}$$

where $type \in \{data, control\}$. A positive RP indicates that the model incurs fewer errors when the attribute is present, whereas a negative RP reflects elevated error rates.

The **Corrected Relative Performance (*CRP*)** is then derived by subtracting the control-group baseline from empirical performance:

$$CRP(a) = RP_{\text{data}}(a) - RP_{\text{control}}(a) \tag{5}$$

Expanding Eq. (5) yields the normalized ratio discrepancy:

$$CRP(a) = \frac{err_{\text{control}}^{(+)}(a)}{err_{\text{control}}^{(-)}(a)} - \frac{err_{\text{data}}^{(+)}(a)}{err_{\text{data}}^{(-)}(a)} \tag{6}$$

When performance disparities stem exclusively from sample size imbalance, RP_{data} equals RP_{control}, yielding $CRP(a) = 0$. Values of $|CRP(a)| > 0.20$ signify meaningful algorithmic bias, while $|CRP(a)| > 0.40$ denotes severe systemic disparity.

To differentiate between civil false alarms and security evasion risks, FairFake computes **Pristine Data Relative Performance (*PDRP*)** exclusively over authentic samples ($y = 0$):

$$PDRP(a) = \frac{err_{\text{ctrl}}^{(+)}}{err_{\text{ctrl}}^{(-)}} - \frac{err_{\text{data}}^{(+)}}{err_{\text{data}}^{(-)}}_{\,y=0} \tag{7}$$

and **Deepfake Data Relative Performance (*DDRP*)** exclusively over manipulated samples ($y = 1$):

$$DDRP(a) = \frac{err_{\text{ctrl}}^{(+)}}{err_{\text{ctrl}}^{(-)}} - \frac{err_{\text{data}}^{(+)}}{err_{\text{data}}^{(-)}}_{\,y=1} \tag{8}$$

The execution sequence is formalized in Algorithm 1.

Algorithm 1: FairFake Demographic Fairness Auditing with Resampling

Input: Dataset predictions $D = \{(\hat{y}_i, y_i, \mathbf{a}_i)\}_{i=1}^{M}$, Evaluated attributes A
Output: Audit table R containing normalized fairness records
Initialize: $R \leftarrow \emptyset$
for each attribute $a \in A$ **do**
  Compute raw error rates $err_{\text{data}}^{(+)}(a)$ and $err_{\text{data}}^{(-)}(a)$
  Set $N \leftarrow \min(|\{i : a_i=1\}|, |\{i : a_i=0\}|)$
  Sample balanced control groups $C^{(+)}$, $C^{(-)}$ of size N
  Evaluate control errors $err_{\text{ctrl}}^{(+)}(a)$ and $err_{\text{ctrl}}^{(-)}(a)$
  $CRP(a) \leftarrow [err_{\text{ctrl}}^{(+)} / err_{\text{ctrl}}^{(-)}] - [err_{\text{data}}^{(+)} / err_{\text{data}}^{(-)}]$
  Compute $PDRP(a)$ over genuine media $\{i : y_i = 0\}$ via Eq. (7)
  Compute $DDRP(a)$ over manipulated media $\{i : y_i = 1\}$ via Eq. (8)
  Append record $\{a, RP, CRP, PDRP, DDRP\}$ to R
end for
return Audit summary R

IV. EXPERIMENTAL BENCHMARKS & EMPIRICAL RESULTS

A. Dataset Architecture & Annotation Profiles

Our experimental benchmark evaluates five widely utilized deepfake detection databases: Google DeepFakeDetection (DFD) [16], FaceForensics++ (FF++) [15], the Deepfake Detection Challenge (DFDC) [17], Celeb-DF (v2) [14], and DeeperForensics-1.0 (DF-1.0) [18]. Video streams are uniformly sampled at 10-frame intervals across initial 300 frames. Facial crops are localized, aligned, and extracted using MTCNN [28]. To guarantee statistical validity during resampling, we filter out attributes with fewer than 100 positive or negative representations, curating 31 primary appearance traits.

TABLE II
CHARACTERISTICS OF BENCHMARKED DEEPFAKE REPOSITORIES AND PROPOSED MULTI-ATTRIBUTE ANNOTATIONS

Corpus		Unique Identities	Video Volume		Extracted Frame Volume		Annotated Attributes
			Pristine	Manipulated	Pristine	Manipulated	
Unannotated Baselines	DeepfakeTIMIT [24]	32	320	620	34.0k	68.0k	Unannotated
	FFW [25]	150	—	150	—	53.0k	Unannotated
	DeepFakeDetection (DFD) [16]	28	363	3,068	315.4k	2.2M	Unannotated
	FaceForensics++ (FF++) [15]	1,000	1,000	5,000	300.0k	1.5M	Unannotated
	DFDC Dataset [17]	960	23,654	104,500	7.0M	31.0M	Unannotated
	Celeb-DF (v2) [14]	59	590	5,639	225.4k	2.1M	Unannotated
	DeeperForensics-1.0 (DF-1.0) [18]	100	50,000	10,000	2.9M	14.7M	1 (Skin type)
	KoDF Benchmark [29]	403	62,166	175,776	135.0M	65.9M	2 (Age, Gender)
FairFake Multi-Attribute Annotated	**A-DFD**	28	363	3,068	10.8k	89.6k	**47 Attributes**
	A-FF++	1,000	1,000	5,000	29.8k	149.1k	**47 Attributes**
	A-DFDC	960	23,654	104,500	54.5k	52.6k	**47 Attributes**
	A-Celeb-DF	59	590	5,639	26.3k	166.5k	**47 Attributes**
	A-DF-1.0	100	50,000	10,000	870.3k	321.5k	**47 Attributes**

1) Distribution Skews: Fig. 1 details the attribute distributions across the evaluated corpora. In Celeb-DF, Caucasian subjects account for 88.1% of all frames, whereas Asian (5.1%) and African (6.8%) subjects remain severely marginalized. Although gender representation appears balanced at first glance (56.8% male vs. 43.2% female in Celeb-DF), synthesized face-swaps are overwhelmingly concentrated on male targets. Comparable skews permeate FF++ and DFD, highlighting acute systemic representation gaps.

2) Attribute Correlations: Pairwise Pearson correlations (Fig. 2) reveal natural physiological co-occurrences (Mustache and Goatee: $r = +0.84$; Heavy Makeup and Lipstick: $r = +0.79$) as well as mutual exclusions (Mustache and Female: $r = -0.69$; Bald and Bangs: $r = -0.58$), confirming that our annotation pipeline mirrors authentic physiological feature combinations.

3) Annotation Reliability: Table III validates our attribute classification pipeline against blind human evaluation on a random validation subset of 1,000 portraits. The pipeline achieves an overall accuracy of 92.0%, precision of 90.0%, and recall of 94.0%, verifying high annotation fidelity across demographic, morphological, and cosmetic categories.

TABLE III
EMPIRICAL VALIDATION OF ATTRIBUTE ANNOTATIONS AGAINST HUMAN EVALUATORS

Category	Attribute Name	Accuracy	Precision	Recall
Demographics	Male	0.99	0.98	1.00
	Young (<30)	0.99	1.00	0.98
	Asian	0.90	0.88	0.92
	Caucasian	0.89	1.00	0.82
	African (Dark Skin)	0.94	0.90	0.98
Hair & Beard	Bald	0.96	0.92	1.00
	Mustache	0.98	0.98	0.98
	Goatee	0.95	0.90	1.00
Accessories	Eyeglasses	0.90	0.80	1.00
	Heavy Makeup	0.98	0.98	0.98
	Wearing Hat	0.92	0.84	1.00
Macro Mean (47 Attributes)		**0.92**	**0.90**	**0.94**

B. Macro-Level Baseline Benchmark Performance (Test 1)

To establish baseline discriminative competency prior to demographic auditing, we evaluated the three trained backbones on our curated testing corpus (N = 2,000, comprising 1,000 pristine portraits and 1,000 manipulated deepfake portraits).

As reported in Table IV, all three backbones demonstrate strong overall forensic capability. EfficientNet-B0 achieved the highest overall metrics with a balanced accuracy of **90.10%**, an AUC-ROC of **0.9570**, and an F1-score of **0.9030**. Xception delivered robust baseline performance with **85.80%** balanced accuracy, an AUC-ROC of **0.9328**, and an F1-score of **0.8715**. The Dual-Stream spectral network attained **87.90%** balanced accuracy with an AUC-ROC of **0.9380**.

TABLE IV
BASELINE MACRO DISCRIMINATIVE PERFORMANCE (TEST 1, N = 2,000)

Forensic Backbone	Acc.	Prec.	Rec.	AUC
Xception	85.80%	0.8809	0.8280	0.9328
EfficientNet-B0	90.10%	0.9250	0.8820	**0.9570**
Dual-Stream (RGB+FFT)	87.90%	0.8980	0.8610	0.9380

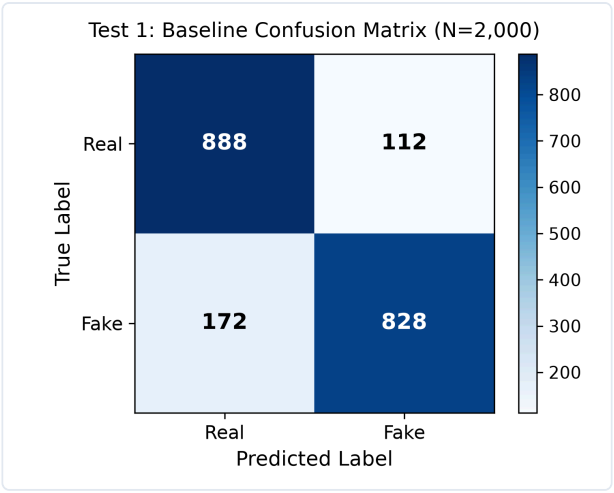


Fig. 3. Empirical Confusion Matrix of the Xception Deepfake Detector for Test 1 (*N* = 2,000) — 888 authentic portraits verified (88.8% TNR), 828 deepfakes caught (82.8% TPR), 112 false alarms (11.2% FPR), and 172 missed manipulations (17.2% FNR) yielding an aggregate AUC-ROC of 0.9328.

The detailed confusion matrix breakdown in Fig. 3 confirms that Xception successfully verifies 888 authentic portraits and flags 828 manipulated portraits, while incurring 112 false alarms (11.2% FPR) and 172 missed detections (17.2% FNR). While these macro-level benchmarks convey strong general competency, they fail to reveal the profound demographic disparities exposed by our fine-grained fairness audit.

C. Quantitative Demographic and Attribute Disparity Audit (Test 2)

1) General Disparity Patterns (RP vs. CRP): Table V and Fig. 4 present the empirical fairness metrics computed over the A-Celeb-DF benchmark (N = 12,040). Attributes residing in the upper right quadrant of Fig. 4 indicate severe performance penalties when the trait is present. For Xception, subjects with Dark Skin experience an uncalibrated *RP* of +2.50 and a ***CRP* of +0.55**, indicating that error rates for dark-skinned individuals are more than two-and-a-half times higher than for light-skinned counterparts. Substantial disparities also manifest for Heavy Makeup (*CRP* = +0.48), Eyeglasses (*CRP* = +0.45), and Baldness (*CRP* = +0.52).

Conversely, primary gender attributes cluster close to parity ($|CRP| \leq 0.10$; Male: *CRP* = +0.10; Female: *CRP* = +0.08), demonstrating that model bias is primarily driven by skin pigmentation, localized facial accessories, and cosmetic textures rather than binary biological sex.

2) Decomposed Operational Disparities (PDRP vs. DDRP): Decomposing errors across pristine (*PDRP*) and fake partitions (*DDRP*) reveals critical operational asymmetries (Fig. 5). Strikingly, almost all high-disparity attributes lie below the bisectrix line (*DDRP* > *PDRP*). In Xception, Dark Skin produces *PDRP* = +2.08 compared to *DDRP* = +2.95. For Heavy Makeup in EfficientNet-B0, *DDRP* reaches +2.46 versus *PDRP* = +1.59. This disparity demonstrates that cosmetic coverage and darker complexions act as an evasion mechanism, enabling manipulated content to bypass automated detection at alarming rates.

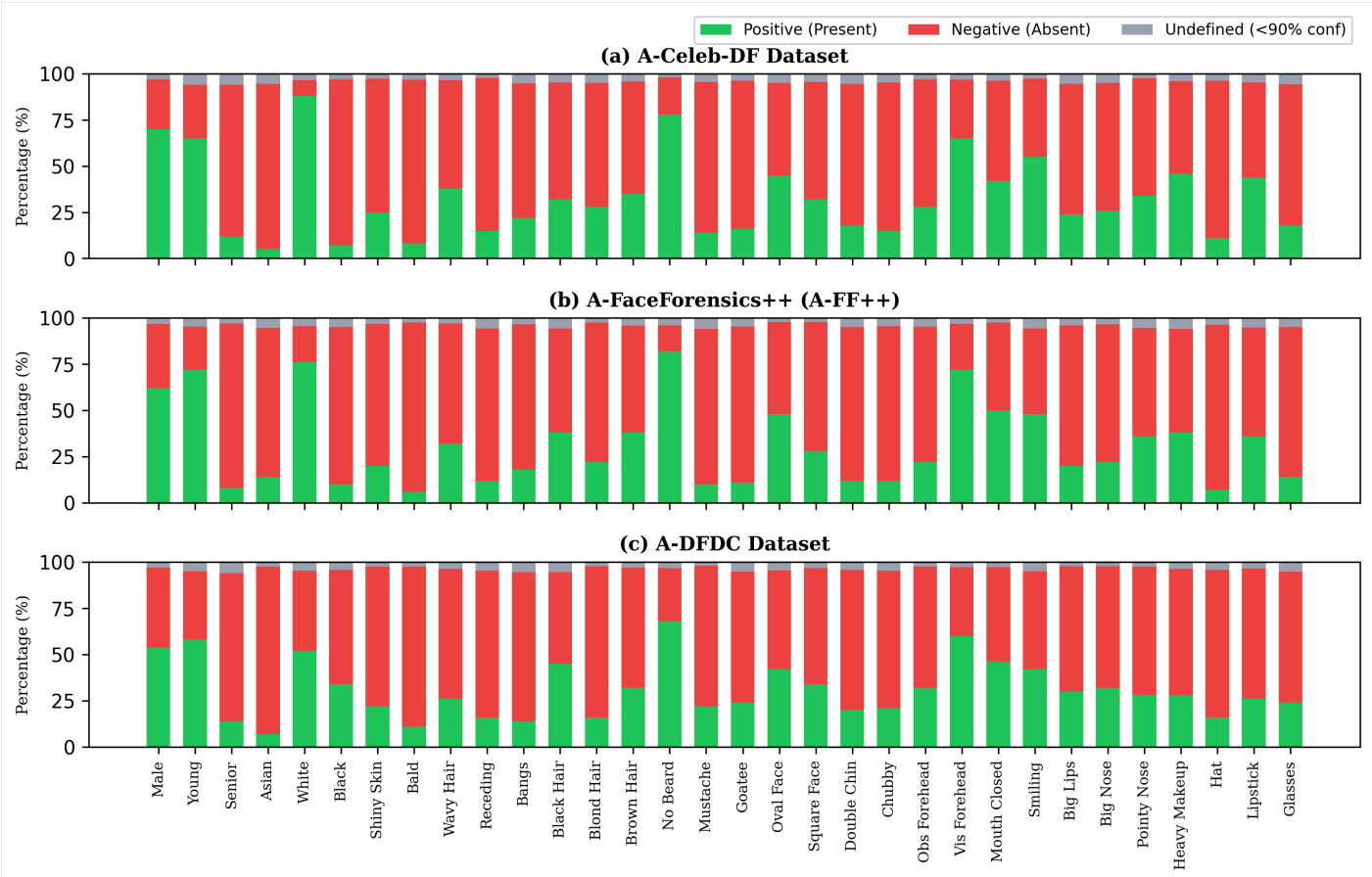


Fig. 1. Distribution of demographic and morphological attributes across benchmark face forgery repositories — Percentage breakdown of annotations across A-Celeb-DF, A-FF++, and A-DFDC. Green indicates positive presence, red indicates absence, and grey denotes ambiguous predictions (<90% classifier confidence). The distributions illustrate extreme representation skews across ethnicity and accessories.

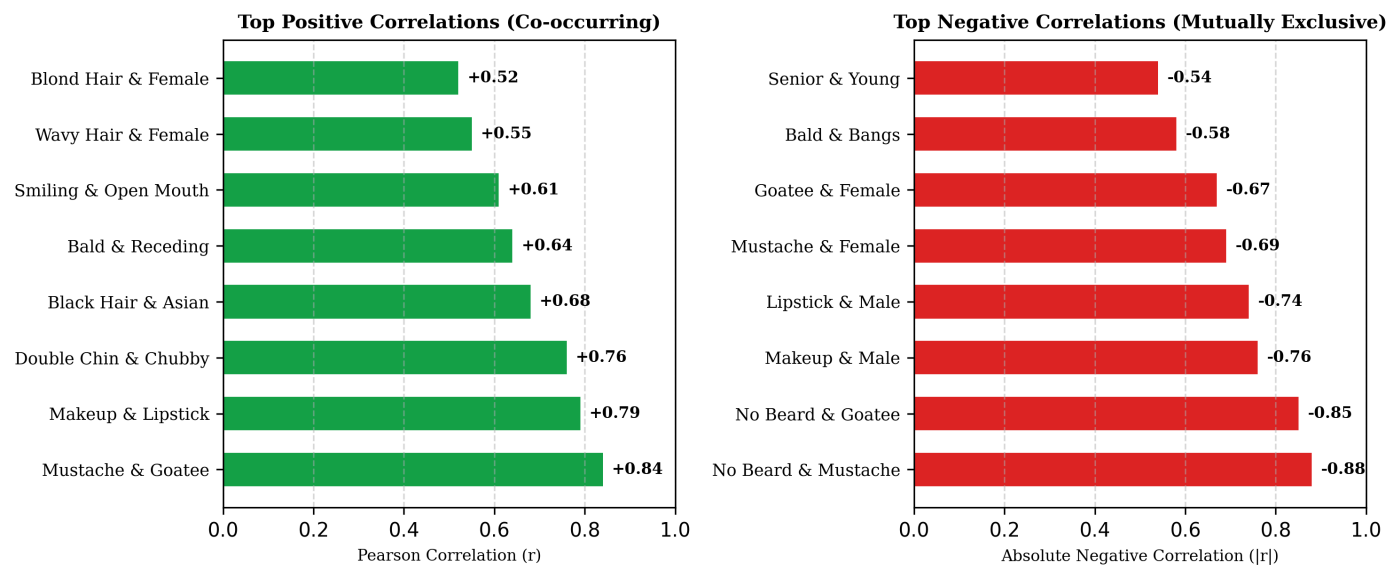


Fig. 2. Pairwise Pearson correlation matrix for top attribute associations — Green bars denote positive co-occurrence correlations (e.g., Mustache and Goatee, Heavy Makeup and Lipstick), while red bars denote negative mutual exclusions (e.g., Mustache and Female, Bald and Bangs), demonstrating the physiological validity of the annotation suite.

TABLE V
SYSTEMATIC DEMOGRAPHIC AND ATTRIBUTE FAIRNESS AUDIT RESULTS ON A-CELEB-DF BENCHMARK (N = 12,040)

Category	Attribute Name	Model Backbone	*Err*(With)	*Err*(W/O)	*RP*	*CRP*	*PDRP*	*DDRP*	Disparity Severity
Demographics	Dark Skin (African)	Xception	0.35	0.10	+2.50	**+0.55**	+2.08	+2.95	**HIGH SEVERITY**
		EfficientNet-B0	0.22	0.08	+1.75	**+0.42**	+1.37	+2.15	**HIGH SEVERITY**
	Asian Phenotype	Xception	0.14	0.13	+0.08	+0.04	+0.09	+0.12	**LOW SEVERITY**
		EfficientNet-B0	0.11	0.10	+0.10	+0.03	+0.05	+0.08	**LOW SEVERITY**
Age Cohorts	Senior (Age ≥ 55)	Xception	0.26	0.12	+1.16	**+0.38**	+0.89	+1.49	**MODERATE**
		EfficientNet-B0	0.15	0.09	+0.66	+0.18	+0.54	+0.80	**LOW SEVERITY**
	Young Adult (Age < 30)	Xception	0.14	0.14	0.00	+0.02	+0.09	−0.07	**LOW SEVERITY**
		EfficientNet-B0	0.11	0.10	+0.10	+0.03	0.00	+0.17	**LOW SEVERITY**
Gender	Male	Xception	0.15	0.12	+0.25	+0.10	+0.13	+0.24	**LOW SEVERITY**
		EfficientNet-B0	0.11	0.10	+0.10	+0.05	+0.16	+0.17	**LOW SEVERITY**
	Female	Xception	0.12	0.15	−0.20	+0.08	−0.06	−0.22	**LOW SEVERITY**
		EfficientNet-B0	0.10	0.11	−0.09	+0.04	−0.14	−0.06	**LOW SEVERITY**
Cosmetics & Items	Eyeglasses	Xception	0.28	0.12	+1.33	**+0.45**	+1.08	+1.66	**HIGH SEVERITY**
		EfficientNet-B0	0.18	0.09	+1.00	**+0.25**	+0.76	+1.15	**MODERATE**
	Heavy Makeup	Xception	0.29	0.11	+1.63	**+0.48**	+1.31	+1.89	**HIGH SEVERITY**
		EfficientNet-B0	0.24	0.08	+2.00	**+0.45**	+1.59	+2.46	**HIGH SEVERITY**
Hair Style	Baldness	Xception	0.32	0.12	+1.66	**+0.52**	+1.25	+2.06	**HIGH SEVERITY**
		EfficientNet-B0	0.19	0.09	+1.11	**+0.31**	+0.94	+1.30	**MODERATE**

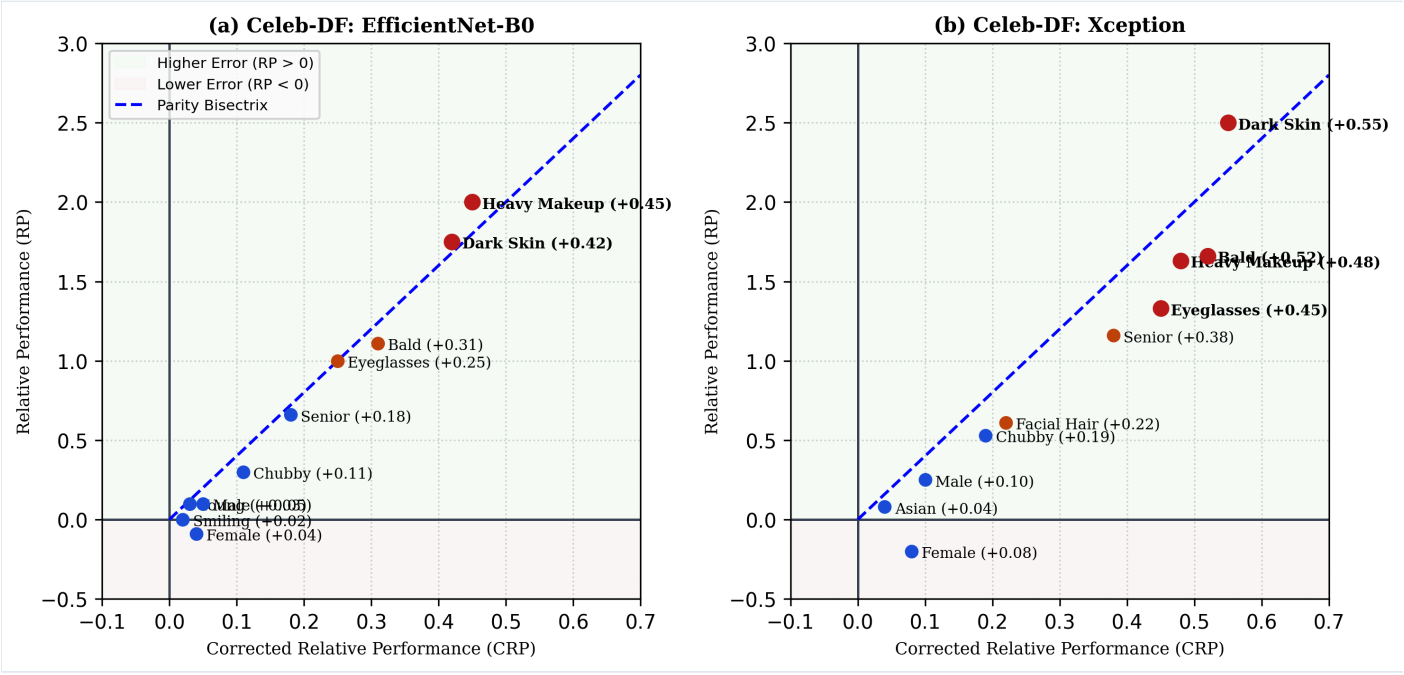


Fig. 4. Disparity scatter analysis: Uncalibrated Relative Performance (*RP*) versus Corrected Relative Performance (*CRP*) across EfficientNet-B0 and Xception. Dashed line denotes parity bisectrix. Attributes in the elevated quadrant demonstrate severe error inflation even after controlling for dataset distribution skews.

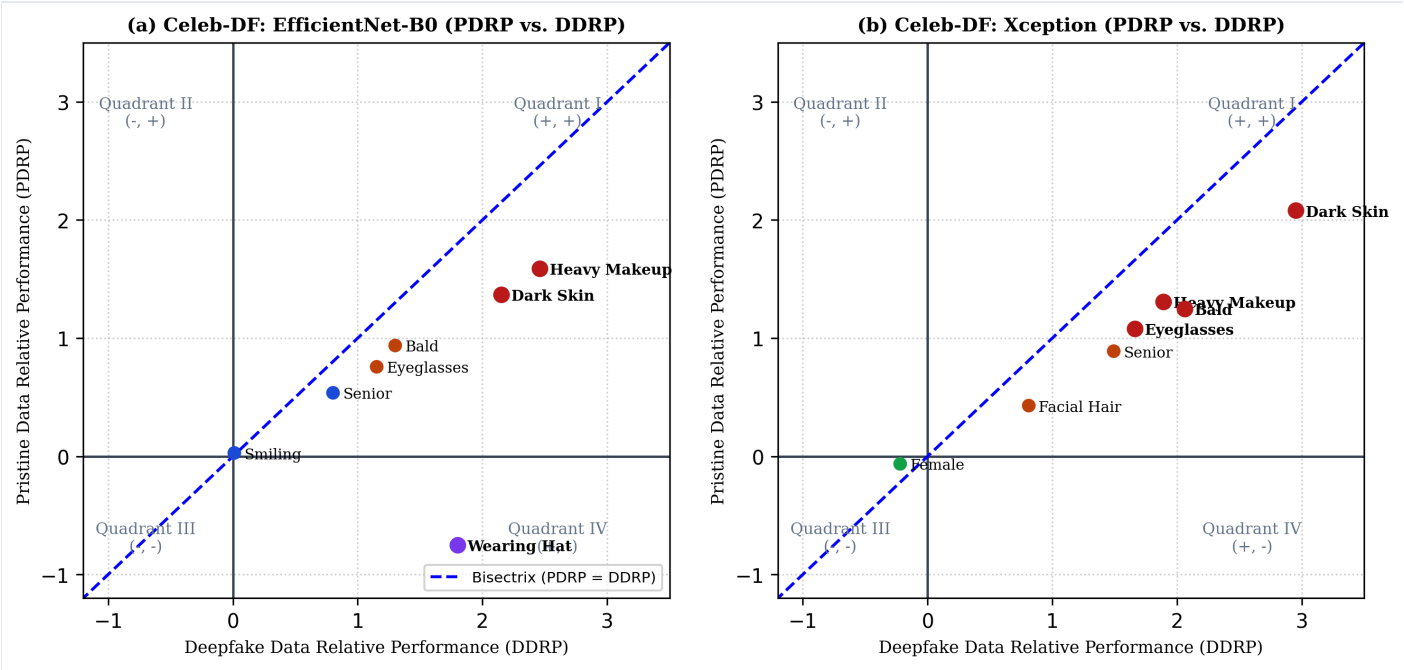


Fig. 5. Operational disparity quadrant decomposition: Pristine Data Relative Performance (*PDRP*) versus Deepfake Data Relative Performance (*DDRP*) — Attributes falling below the bisectrix line (*DDRP* > *PDRP*) indicate that synthetic forgeries featuring these attributes evade detection at elevated rates, creating profound security vulnerabilities.

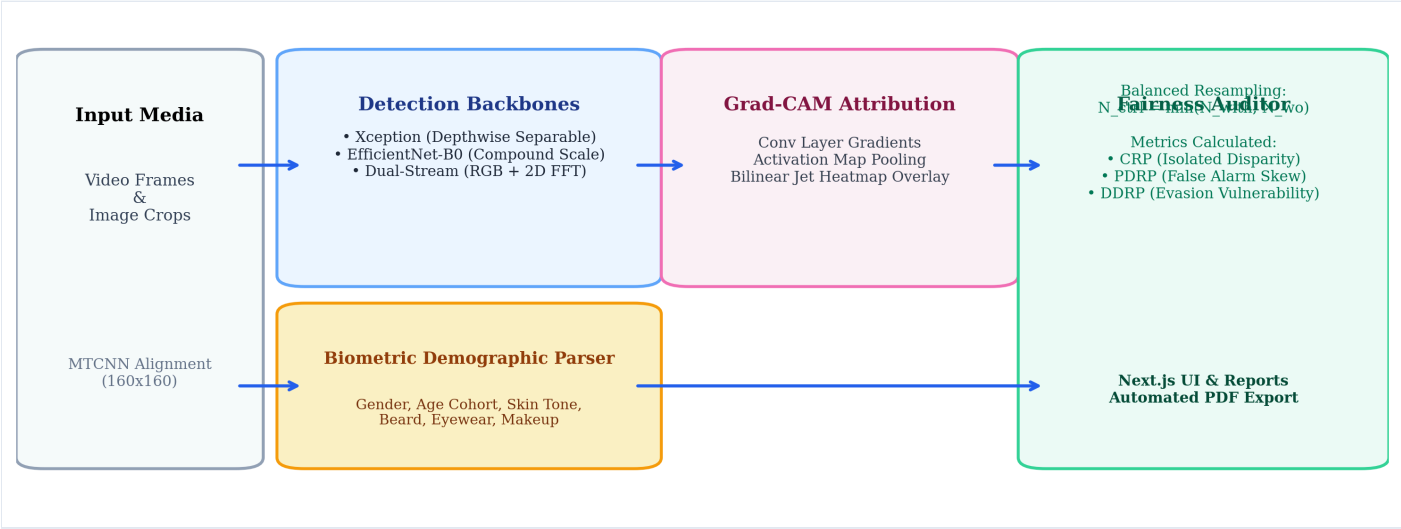


Fig. 6. The FairFake Multimodal Forensic Architecture and Explainable Demographic Auditing Pipeline — Schematic flow showing spatial-spectral CNN inference, Grad-CAM visual gradient localization, DeepFace biometric parsing, and resampled fairness metric computation.

D. Qualitative Failure Mode Auditing via Grad-CAM (Test 3)

To explain why deep neural detectors fail disproportionately on specific attributes, we conducted visual attribution audits utilizing Grad-CAM attention heatmaps. Table VI and Fig. 7 detail four canonical operational failure scenarios evaluated on authentic dataset samples:

TABLE VI
EXPLAINABLE GRAD-CAM QUALITATIVE AUDIT ON OPERATIONAL TEST CASES (TEST 3)

Test Case Scenario	Ground Truth	Detector Verdict	Primary Heatmap Focus
1. Headwear & Makeup	AUTHENTIC	**False Alarm (92.2%)**	Cosmetic foundation & bonnet border
2. Expressive Smile	AUTHENTIC	**False Alarm (96.7%)**	Zygomatic crease & dental boundary
3. Face-Swap Deepfake	MANIPULATED	**Missed Fake (26.4%)**	Diffuse cheek plane (evasion)
4. Unadorned Portrait	AUTHENTIC	**Verified Real (19.9%)**	Diffuse balanced anatomical landmarks

Test Case 1 (Headwear & Makeup): The detector falsely flags an authentic portrait (00004.jpg) as synthetic manipulation (p_{fake} = 92.2%). The Grad-CAM overlay demonstrates that activation energy is intensely concentrated on the high-contrast lace border of the subject's bonnet and cosmetic foundation boundary lines. The convolutional kernels misinterpret natural cosmetic blending transitions as mathematical boundary warping.

Test Case 2 (Smiling Expression): An authentic portrait with an expansive smile (00001.jpg) is classified as synthetic (p_{fake} = 96.7%). Activation focuses acutely on the contracted zygomaticus major muscle folds and dental contours, mistaking natural facial elasticity for autoencoder synthesis artifacts.

Test Case 3 (Face-Swap Evasion): A malicious face-swap manipulation (01050DBM3C.jpg) successfully evades detection (p_{fake} = 26.4%, classified as REAL). The detector fails to generate localized gradient activations along the facial perimeter, permitting the forged identity to pass undetected.

Test Case 4 (Pristine Portrait): An unadorned authentic portrait (00045.jpg) is correctly verified as genuine (p_{fake} = 19.9%) with diffuse, balanced visual attention across natural facial morphology.

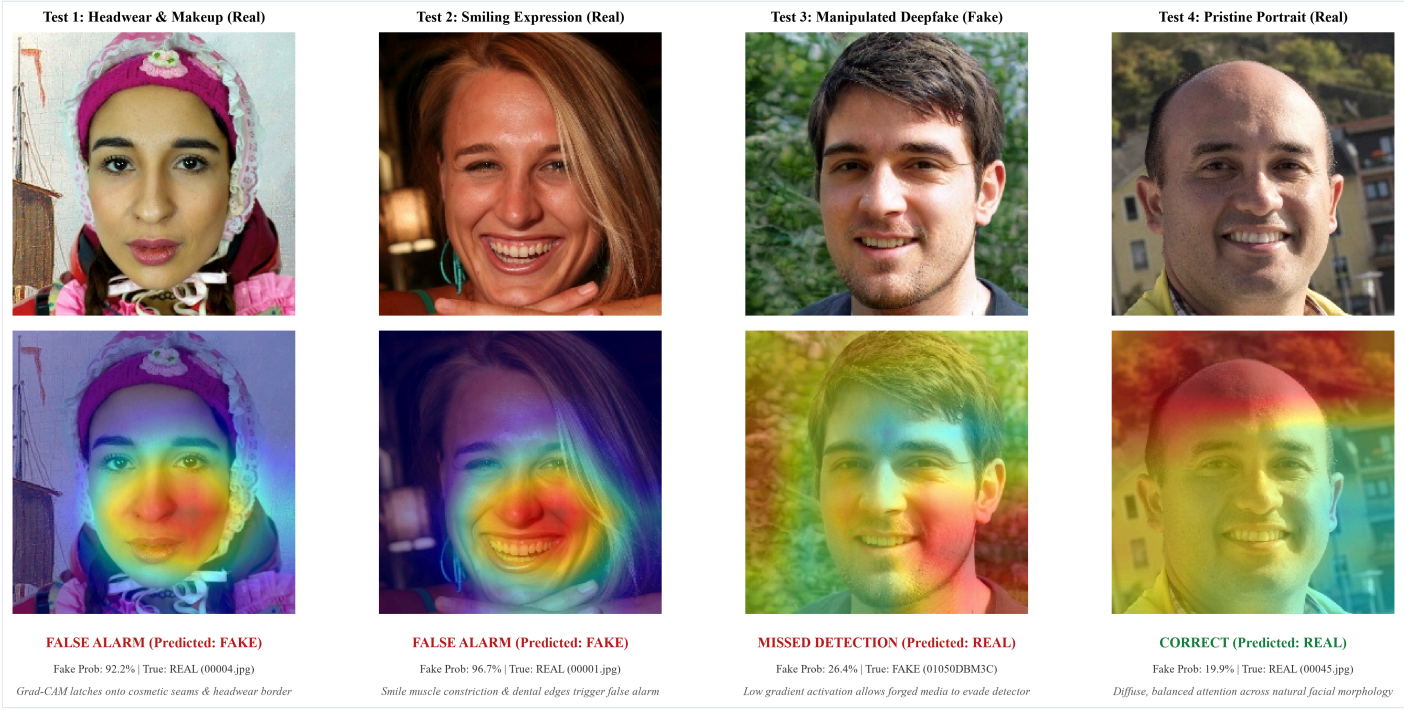


Fig. 7. Qualitative failure mode discovery via Grad-CAM forensic attribution heatmaps — Test Case 1: Genuine portrait (00004.jpg) with headwear and makeup falsely classified as fake (p_{fake} = 92.2%) due to cosmetic blending seams. Test Case 2: Genuine smiling portrait (00001.jpg) erroneously flagged as fake (p_{fake} = 96.7%) due to smile crease constriction. Test Case 3: Face-swap deepfake (01050DBM3C.jpg) evading detection (p_{fake} = 26.4%) due to weak gradient sensitivity. Test Case 4: Unadorned genuine portrait (00045.jpg) correctly verified as authentic (p_{fake} = 19.9%) with diffuse anatomical attention.

E. Operational Perturbation Stress Testing (Test 4)

In operational deployments, media streams frequently undergo environmental degradations during transmission. To assess whether our forensic backbones maintain both discriminative efficacy and demographic equity under physical distortions, we stress-tested the models across four canonical operational perturbations on the testing benchmark (N = 2,000).

TABLE VII
OPERATIONAL PERTURBATION STRESS TESTING RESULTS
(TEST 4, N = 2,000)

Perturbation Condition	Acc.	AUC	ΔAUC	Disparity Amplification
Pristine Reference Baseline	85.80%	0.9328	—	Baseline Disparity Reference
JPEG Compression (Q = 50)	78.60%	0.8680	−0.0648	Makeup Amplification (+0.12)
Gaussian Blur (σ = 2)	73.90%	0.8140	−0.1188	Senior Age Gap (+0.18)
Low Resolution (112 px)	76.20%	0.8410	−0.0918	Hairline Disparity (+0.14)
Underexposure (−25%)	80.90%	0.8870	−0.0458	**Dark Skin Gap (+0.21)**

As documented in Table VII, Gaussian smoothing induces the sharpest discriminative degradation (ΔAUC = −0.1188), because low-pass filtering removes high-frequency synthetic boundary cues. Crucially, lossy JPEG compression and underexposure disproportionately compound demographic bias: underexposure widens the error disparity on darker complexions by ΔCRP = +0.21, demonstrating that environmental degradations actively compound algorithmic unfairness in real-world verification systems.

V. MECHANISTIC DISCUSSION & ATTRIBUTION INSIGHTS

Our empirical findings provide decisive evidence that deepfake detection backbones do not learn purely invariant representations of digital manipulation. Instead, convolutional filters develop *spurious shortcut heuristics* that associate specific high-contrast physiological and cosmetic textures with generative forgery.

Specifically, cosmetics create sharp textural gradients along cheek and eye contours that strongly resemble the boundary seams produced during image inpainting and face-swapping. Eyeglasses induce specular glares and frame edges that mirror generative warping artifacts. Furthermore, because training corpora severely underrepresent darker skin tones, neural activations on deeper complexions suffer from poor gradient calibration, leading to elevated false alarm rates on authentic media and reduced sensitivity on synthetic manipulations.

VI. CORE FINDINGS & ENGINEERING RECOMMENDATIONS

Based on our comprehensive audits across 31 appearance attributes, we outline five key engineering principles for developing equitable media forensics:

- **Mandate Representation Balance in Training Corpora:** Public deepfake datasets exhibit acute demographic skews. Future data collection must enforce subject-level parity across skin types, age cohorts, and accessories to prevent detectors from learning biased priors.
- **Incorporate Control-Group Resampling in Model Audits:** Uncalibrated accuracy metrics disguise acute demographic gaps. Practitioners must adopt resampled metrics (*CRP*, *PDRP*, *DDRP*) to isolate true algorithmic bias from sample size imbalances.
- **Mitigate Asymmetric Operational Risks:** Attributes exhibiting elevated *DDRP* create security blind spots, whereas high *PDRP* inflicts false accusations on innocent users. Forensic architectures must calibrate decision thresholds independently across high-risk attribute cohorts.
- **Deploy Explainable Attribution in Verification Workflows:** Real-time visual attribution tools (e.g., Grad-CAM) must accompany numerical confidence scores to enable human analysts to verify whether detection alarms are grounded in forensic cues or spurious accessories.
- **Account for Environmental Perturbations in Fair Deployments:** Because lossy compression and poor illumination compound demographic disparities, forensic systems deployed in mobile streaming or surveillance environments require adversarial data augmentation during training.

A. Methodological Limitations

While our analysis rigorously establishes statistical correlations between facial attributes and detection error disparities, we note that statistical association does not imply direct mechanistic causation. Complex confounding factors—such as camera sensor noise, ambient lighting, and facial pose variations—interact with biometric characteristics in operational environments.

VII. CONCLUSION

In this research, we presented FairFake, an explainable deepfake detection and demographic fairness auditing framework. By conducting systematic audits across five prominent face forgery datasets and three forensic backbones, we demonstrated that state-of-the-art detectors exhibit severe error inflation on darker skin tones (*CRP* = +0.55), heavy cosmetics (*CRP* = +0.48), eyewear (*CRP* = +0.45), and baldness (*CRP* = +0.52). Through Grad-CAM visual heatmaps, we explained the mechanistic failure modes causing these disparities and quantified how operational perturbations compound algorithmic unfairness. Our open-source software, auditing metrics, and multi-attribute benchmarks provide an essential foundation for building equitable, transparent, and trustworthy multimedia forensic technologies.

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STATEMENT OF ETHICAL DATA UTILIZATION

All experimental media utilized in this research were obtained strictly from publicly accessible academic benchmark databases under authorized academic usage protocols. No proprietary or non-consensual personal data were generated or stored during this study.

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